

Practice Questions - by Reza Sir

Question 1

X	Y
2	14
4	29
6	42

-If you fit the above data samples using $Y=mx+b$, where $m = 3$ and $b = 1$, calculate the amount of error using mean square error loss function.

-Perform one iteration of gradient descent to update both m and b , assume that the learning rate is 0.01

Question 2

-Describe the differences between regression and classification.

-What should you do regarding the learning rate if you are close to the global minima while your model performance is fluctuating up and down?

-In the mean square error loss function, why do we omit the negatives by squaring it, instead of using absolute?

-Why entropy is a better loss function than mean square error for classification problems?

Question 3

X	Y
0.2	0
0.4	0
0.8	1

-If you fit the above data samples using $Y=mx+b$, where $m = 3$ and $b = 1$, calculate the amount of error using binary crossentropy loss function.

-Perform one iteration of gradient descent to update both m and b , assuming that the learning rate is 0.1

Question 4

	Y	!Y
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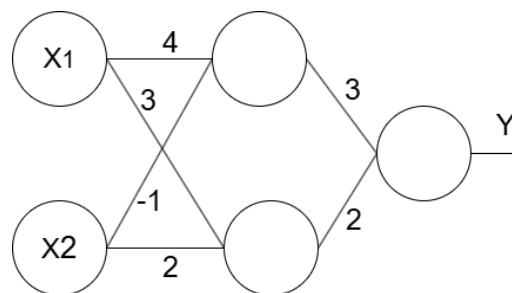
	Z	!Z	Z	!Z
X	.246	.121	.214	.006
!X	.024	.146	.008	Q

1. Find out the value of Q
2. Calculate $P(!X \text{ and } !Z)$

Question 5

Mr. Z has bought a lie detector machine. The machine can properly detect lie 97% of the time and properly detect truth 6% of the time. If stats say that around 5% people are liar in general, and if the lie detector machine catches a person telling lie, find out the probability of the person speaking the truth.

Question 6



- If $X1 = 5$ and $X2 = 1$, apply forward propagation to calculate the value of Y. Consider that the hidden layer uses ReLU activation function while the output layer uses the sigmoid activation function.
- Mention the importance of activation functions in a neural network.

Question 7

X	Y	Z
a	b	c
a	!b	c
!a	!b	c
!a	!b	c
a	!b	!c
a	b	!c
a	b	!c
a	b	!c

- Considering three variables X, Y, and Z, construct a joint distribution table from above.
- Calculate $P(!a | !b)$

Question 8

	Left-handed		Right-handed	
	Male	Female	Male	Female
Pass	.019	.024	X	.223
Fail	.112	.011	.014	.324

Consider that the following joint distribution table has been constructed from three variables:

Gender = {Male, Female}, Result = {Pass, Fail}, Writing = {Left-handed, Right-handed}.

- From the table above, calculate $P(\text{Male})$
- What is $P(\text{Pass and Fail})$?
- Determine if Gender and Result are conditionally independent of each other given Right-handed.

Question 9

-Mr. X bought a lie detector machine which can detect lie in 97% of the cases and cannot detect truth in 5% of the cases. If 7% people visiting Mr. X are liars, and in case his detector machine detects someone as a liar, what is the probability of the person speaking the truth?

Question 10

- Discuss the ways to minimize the zero probability issue in Naive Bayes.

F1	F2	F3	Label
a	x	m	Yes
a	x	m	No
a	y	n	No
y	y	n	Maybe
a	y	n	No
b	x	n	Maybe
a	x	m	Maybe
b	y	n	No
b	y	n	Yes

a	x	n	Yes
a	x	n	No
b	x	n	Yes
b	y	n	Maybe

-Determine if the decision should be 'Yes' or 'No' given F1=b, F2=y, and F3=m. If you encounter zero probability, take any of the necessary steps to minimize it.

Question 11

-For a dataset of 10 samples with binary (Yes, No) labels, draw a graph of entropy against the label attribute distribution ratios.

N.B. label attribute distribution ratios: if there are 1 yes and 9 no, the ratio is 1/9. Similarly, if there are 3 yes and 9 no, the ratio is 3/9. The X axis of the graph should represent the ratios, while the Y axis should represent the corresponding entropy value

F1	F2	F3	Label
a	x	m	Yes
a	x	m	No
a	y	n	No
b	y	n	Maybe
a	y	n	No
b	x	n	Maybe
a	x	m	Maybe
b	y	n	No
b	y	n	Yes
a	x	n	Yes
a	x	n	No
b	x	n	Yes
b	y	n	Maybe

-Calculate the information gain for F1 and F2, determine which one is the better feature for a decision tree.

-Supposedly, if the root node of a decision tree represents feature F3, on its 'n' branch, determine the information gain for F1 and F2.

-In the above table, is it possible to have the maximum possible entropy value (In case of three labels) for the label column if you are allowed to change the label distribution? Why or why not?

Quiz 3 and 4 Question

Mood	Day	Genre	FriendSuggested	Watched
Happy	Weekend	Comedy	Yes	Yes
Tired	Weekday	Action	No	No
Excited	Weekend	Horror	Yes	Yes
Sad	Weekday	Drama	No	No
Happy	Weekday	Comedy	Yes	Yes
Excited	Weekend	Action	Yes	Yes
Tired	Weekday	Comedy	No	No
Sad	Weekend	Drama	Yes	Yes
Happy	Weekday	Horror	No	No
Excited	Weekend	Comedy	Yes	Yes

a. A person is trying to decide whether to watch a movie. Here are the details:

- They are feeling **excited**.
- It's the **weekend**.
- The movie is a **horror** film.
- A friend has **not suggested** the movie.

Based on the dataset, **will the person watch the movie?**

Use either a **Naive Bayes classifier** to justify your prediction.

B. Create a decision tree using ID3